

Are Medical Agents Reliable Co-workers for Radiologists?

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Radiologists face a significant burden

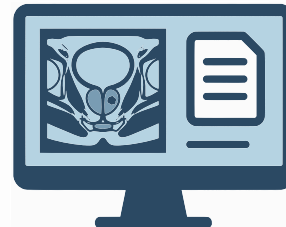
Clinical Workflow



Image Acquisition
(mp-MRI)



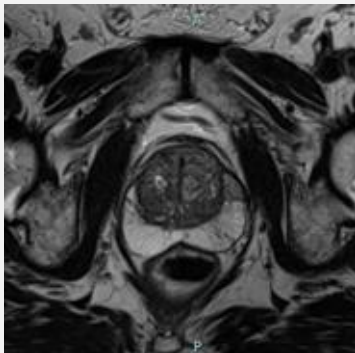
Data Storage
(Secure Network)



Viewing
(Workstation)



Decision-making
(Radiologist)



Radiologists diagnosis ~20 MRI cases/day;

Each case takes ~25 mins;

> 8h of intense focus.

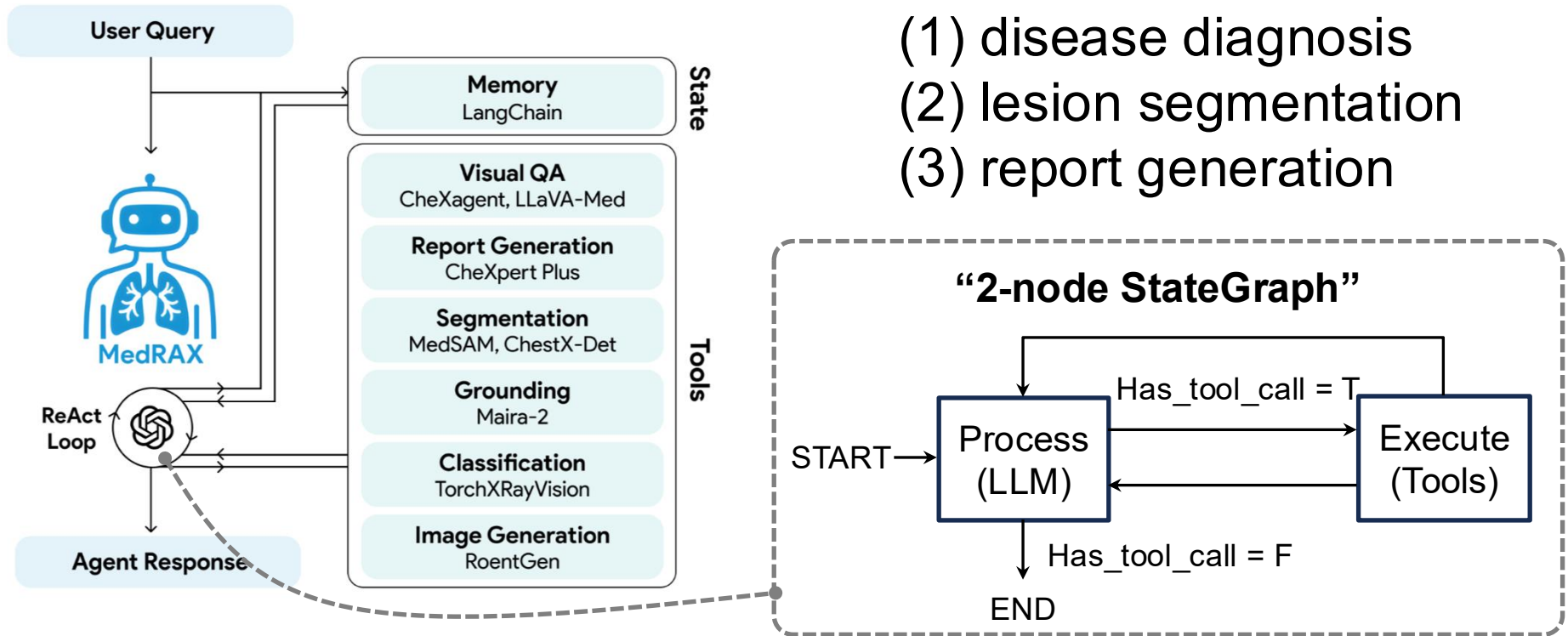
A co-worker largely reduces their loads

“When working with a radiologist trainee who has pre-processed the case by conducting preliminary analysis and segmentation. I can finalize a case in **5 minutes.**”



Dr. Oguz Akin,
Radiologist (>15 years of experience),
Memorial Sloan Kettering Cancer Center

Medical agents are increasingly capable



Are they reliable clinical “co-workers”?

Our intuition

To be a reliable “co-worker”, the medical agent should be more than a tool hub. It should also be able to:

- (1) adapt to ***new skills quickly***,
- (2) communicate in ***radiologist-aligned language***,
- (3) and admit when it ***does not know***.

A reliable clinical “co-worker”

Project A

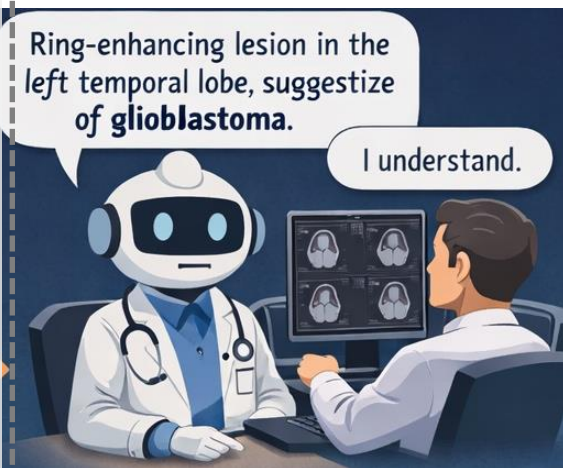
Rapid Skill Acquisition



Keeps pace with rapidly evolving medical tools and protocols.

Project B

Seamless Clinical Dialogue



Communicates using standard medical terms and report structures.

Recognizing Their Limits



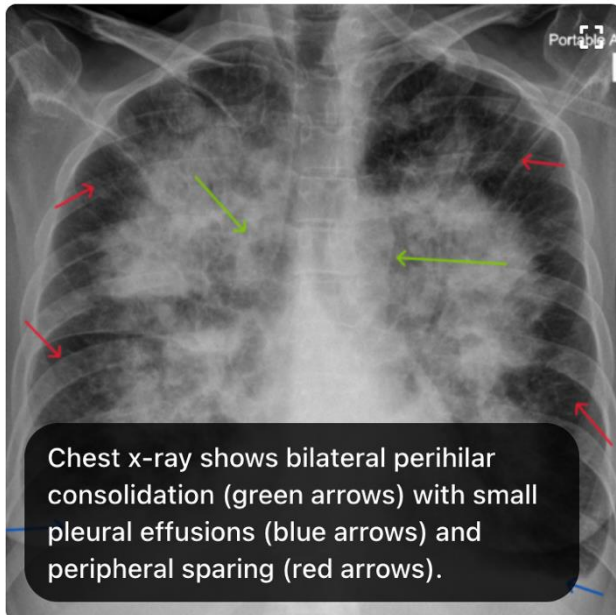
Acknowledges uncertainty to avoid misdirecting the radiologist.

We evaluate three pillars of clinical reliability: skill acquisition, communication clarity, and unknown acknowledgement.

Project B: Dataset curation

Expert diagnosis workflow:

history → imaging → differential → diagnosis



CLINICAL HISTORY

A 51-year-old woman with a 23-year history of adult-onset asthma and allergic rhinitis, requiring daily bronchodilator and corticosteroid use, presented with several days of diarrhoea, abdominal pain, and dyspnoea. Initial examination revealed bilateral wheeze and mild lower limb petechiae. Laboratory testing demonstrated marked leukocytosis with a peripheral eosinophilia percentage of

IMAGING FINDINGS

A portable chest x-ray obtained in the Emergency Department demonstrated extensive bilateral perihilar airspace consolidation with small bilateral pleural effusions and relative sparing of the peripheral lung fields (Figure 1).

DISCUSSION

Background

EGPA is an uncommon systemic vasculitis that typically occurs in three sequential phases: a prodromal phase involving asthma attacks and/or allergic rhinosinusitis, an eosinophilic phase characterised by peripheral eosinophilia and

Outcome

The patient developed acute hypoxic respiratory failure requiring intensive care admission and intubation. High-dose corticosteroid therapy led to rapid clinical and haematological improvement, with eosinophil counts normalising within three days. She was successfully extubated and continues to undergo multidisciplinary follow-up.

Project B: Dataset curation

Fruits

case_id: 19487

title: Thoracic Kaposi sarcoma

patient_age: 22

patient_gender: male

clinical_history: A 22-year-old male with a 2-year history of HIV, previously poorly compliant on highly active antiretroviral therapy (HAART), presents with recurrent episodes of dyspnoea, fatigue, haemoptysis, and facial swelling. He was previously treated for Pneumocystis jirovecii pneumonia (PJP), a bacterial pneumonia.

images: [8 items]

image_descriptions: [8 items]

gt_imaging_findings: CT lung window shows bilateral nodular regions of consolidation with a peribronchovascular distribution and characteristic flame-shaped appearance, most severe in the lower lung zones. There is also bilateral nodular thickening of the pleura and small pleural effusions. The soft tissue windows show ill-defined and infiltrative soft tissue in the axilla, mediastinum and hilar regions, with poorly defined lymphadenopathy. There are also multiple regions of skin thickening, such as in the anterior left chest wall. Chest x-ray revealed worsening of the perihilar reticulonodular opacities in the mid and lower lung zones.

gt_differentials: [6 items]

gt_final_diagnosis: Thoracic Kaposi sarcoma

images[0]

eurorad_dataset/pdf_images/19487_fig_1_0.png

images[1]

eurorad_dataset/pdf_images/19487_fig_1_1.png

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images[4]

eurorad_dataset/pdf_images/19487_fig_unk_4.png

images[5]

eurorad_dataset/pdf_images/19487_fig_3_5.png

images[6]

eurorad_dataset/pdf_images/19487_fig_3_6.png

images[7]

eurorad_dataset/pdf_images/19487_fig_4_7.png

Statistics: 234 cases collected.

Project B: Evaluation setup

Dimension	Clinical Communication	Acknowledge Unknown
<i>Purpose</i>	Measure clinical reasoning quality	Measure unknown with missing information
<i>Input data</i>	Eurorad cases	
<i>Variants per case</i>	1: Complete	4: complete, no history, no image, no both
<i>Output schema</i>	4-step JSON: history → findings → differentials (top-5) → final dx + confidence + reasoning	
<i>Agent brain</i>	gpt-4o/Qwen-3VL (8B)	
<i>Metric computation</i>	LLM-as-a-judge (gpt-5.4)	
<i>Interactive mode</i>	Static (offline)	

Project B: Evaluation metric (clinical)

Metric	What it measures												
<i>Imaging Findings Score</i>	GPT-judged semantic match of findings to GT (0–1)												
<i>Final Diagnosis Score</i>	GPT-judged semantic match of dx to GT (wrong/same family/same)												
<i>Differential Recall@1 / @3</i>	Did GT dx appear in top-1/-3 of differentials?												
<i>Consistency</i>	Do stated findings logically support stated dx? (0-1)												
<i>Communication Quality</i>	1.0 expert · 0.5 adequate · 0.0 unhelpful												
<i>Reasoning Coherence</i>	<table><tr><th>Confidence</th><th>Correct (dx ≥ 0.5)</th><th>Wrong (dx < 0.5)</th></tr><tr><td>High</td><td>1.0 confident & right</td><td>0.0 dangerous overconfidence</td></tr><tr><td>Medium</td><td>0.8 correct but hedged</td><td>0.4 wrong & uncertain (marginal)</td></tr><tr><td>Low</td><td>0.5 correct but too conservative</td><td>1.0 wrong but knew it (epistemic honesty)</td></tr></table>	Confidence	Correct (dx ≥ 0.5)	Wrong (dx < 0.5)	High	1.0 confident & right	0.0 dangerous overconfidence	Medium	0.8 correct but hedged	0.4 wrong & uncertain (marginal)	Low	0.5 correct but too conservative	1.0 wrong but knew it (epistemic honesty)
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<i>Uncertainty Appropriateness</i>													
<i>Reliable Co-worker Score</i>	Avg (communication, reasoning, uncertainty)												

Project B: Evaluation metric (unknown)

Metric	What it measures
<i>Overconfidence Rate</i>	% cases rated "high" confidence when info missing
<i>Uncertainty Rate</i>	% cases rated "low"/"medium" confidence
<i>Hedging Rate</i>	Output contains explicit uncertainty language (regex)
<i>Confidence Drop vs. complete</i>	% cases where confidence fell when info was withheld

Project B: Result I

Table 1: **Clinical Communication.** RCW = Reliable Co-worker Score = avg(communication, reasoning coherence, uncertainty). Dx = Diagnosis. Higher is better for all metrics

Model	Mode	<i>N</i> *	Findings	Dx@1	Dx@3	Final Dx	Consist.	RCW
gpt-4o	baseline	32	0.268	25.0%	25.0%	0.227	0.782	0.466
	agent	32	0.236	25.0%	28.1%	0.222	0.762	0.393
Qwen3-VL (8B)	baseline	32	0.183	6.2%	18.8%	0.119	0.746	0.406
	agent	32	0.201	9.4%	12.5%	0.137	0.708	0.378

*Evaluation is performed on 32 CXR cases; scaling to the full collected dataset is left for future work due to budget constraints.

The signature pattern:

- Agent reason like a junior trainee whose **logic is sound** but whose **eye is undertrained**.

Benefits of agent:

- Qwen3-VL + tools = better observation (Findings 0.183 $\rightarrow$ 0.201, top-1 dx 6.2% $\rightarrow$ 9.4%).
- GPT-4o + tools = no eyesight gain (Already strong enough natively).
- Consistency stays roughly same (Agents see more but think the same way).

Project B: Result II

Table 2: **Acknowledge Unknown.** Conf Drop% measures the relative reduction in high-confidence outputs vs. the complete-information baseline.

Model	Variant	<i>N</i>	Final Dx	High%	Low / Med%	Hedged%	Conf Drop%
gpt-4o	complete	32	0.297	59.4%	40.6%	3.1%	—
	no_history	32	0.106	53.1%	46.9%	68.8%	34.4%
	no_image	32	0.196	6.2%	93.8%	100.0%	56.2%
	no_both	32	0.003	0.0%	100.0%	100.0%	96.9%
Qwen3-VL (8B)	complete	32	0.216	81.2%	18.8%	0.0%	—
	no_history	32	0.037	59.4%	40.6%	37.5%	25.0%
	no_image	32	0.139	6.2%	93.8%	100.0%	81.2%
	no_both	32	0.000	0.0%	100.0%	100.0%	96.9%

Findings:

- **Qwen3-VL** is much more overconfident.
- **GPT-4o** is better-calibrated.
- Image absence is salient; history absence is invisible.
- A calibration gap most pronounced in the smaller model.

Models recognize when they can't see the patient (X-ray), but not when they don't know the patient (clinical history).

Project B: Demo



Medical Reasoning Agent for Chest X-ray

Upload a chest X-ray (JPEG / PNG / DICOM) and ask me anything about it.

Tip: drag a file into the chat box or click the attachment icon.

Type your message here...



Takeaway & Future work

Q: Are Medical Agents Reliable Co-workers for Radiologists?

A: Not yet.

Add self-reflection

- Try one round of “review your draft; remove contradictions; check tool evidence supports each finding.”

Scale evaluation

- Scale to $N \geq 200$.
- Multi-judge ensemble.

Scope of work

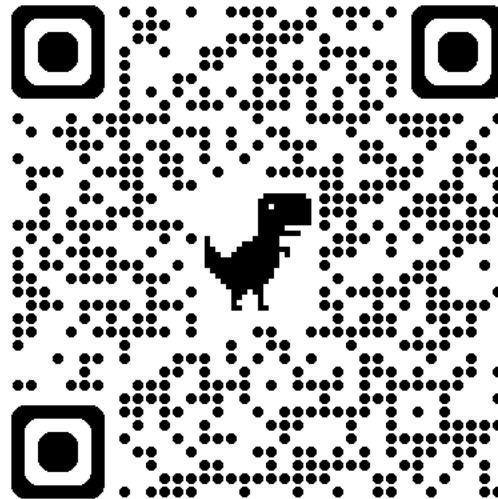
Mengmeng Ma (Project lead)

- Conceived and designed the overall approach;
- Implemented all components used in the final deliverable, including the production UI and dataset construction.
- Conducted all experiments, prepared the presentation materials, delivered the demo, and managed the GitHub repository and pull requests.

Kathleen O'Donovan

- Supported raw data collection from Eurorad;
- Contributed to an early Gradio UI prototype (not included in the final demo).

Thanks



Video for this presentation and demo.

team-07 final #14

View status

Code 

 Open [mengmenm](#) wants to merge 1 commit into [Qdata4Capstone:main](#) from [mengmenm:main](#) 

 Conversation **0**

 Commits **1**

 Checks **0**

 Files changed **27**

+5,586 -391 



mengmenm commented [12 minutes ago](#) 

No description provided.



[team-07 final](#)

[ff01399](#)

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one assigned

Labels